

Annamalai Kalyanasundaram

Full-stack GenAI Engineer · RAG & Agentic AI · Python / FastAPI · AWS Certified Solutions Architect

Bengaluru, India · (+91) 7838971194 · annamalai.k97@hotmail.com · linkedin.com/in/annamalai-kalyanasundaram · github.com/<your-github>

PROFESSIONAL SUMMARY

Full-stack GenAI engineer with ~2 years delivering production software and cloud systems, specializing in agentic LLM applications for regulated workflows. Build end-to-end GenAI services spanning LangGraph/LangChain/CrewAI orchestration, Retrieval-Augmented Generation (ingestion, embeddings, vector search, grounded generation), FastAPI microservices, Docker/Kubernetes packaging, and evaluation/observability with RAGAS and LangSmith. Strong foundation in Python, REST API design, AWS (EC2, VPC, IAM, Lambda, ECS), CI/CD, and Git-based collaboration. AWS Certified Solutions Architect – Associate. Seeking Full-stack GenAI Analyst II roles shipping practical GenAI into risk and model-operations environments.

TECHNICAL SKILLS

Generative AI & LLMs: Transformer architectures (self-attention, multi-head, cross-attention), prompt engineering & guardrails, Hugging Face, PyTorch, multimodal LLMs, Ollama, RAG & Advanced RAG (Corrective RAG, Self-RAG, GraphRAG, hybrid search), vector databases (FAISS, ChromaDB, Pinecone, Milvus), RAGAS evaluation, LangChain, LangGraph, CrewAI, agentic AI, MCP, A2A, LangSmith observability, Neo4j (Cypher), PEFT (LoRA/QLoRA), quantization, model serving (vLLM, TensorRT-LLM)

Backend & APIs: Python, FastAPI, REST microservices, request validation, structured error handling, Node.js, Express.js

ML / NLP: NLP fundamentals, embeddings, Transformer architectures, PyTorch

Cloud & DevOps: AWS (EC2, VPC, S3, IAM, CloudFront, RDS, Route 53, CloudWatch, Lambda, Auto Scaling, ELB, ECS), Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins), Git, Azure AD

Data & UI: MySQL, SQL optimization, document ingestion for RAG; React.js; Kotlin / Jetpack Compose / Android SDK

PROFESSIONAL EXPERIENCE

Software Developer — Arche Global Pvt Ltd Bengaluru · Jun 2024 – Jul 2026

- Enhanced **Pulse**, a hybrid-cloud operations platform, driving a ~30% improvement in operational efficiency through automated monitoring and streamlined workflows.
- Designed and maintained **15+ RESTful API endpoints** (Node.js, Express.js) for asset management, ticketing, and monitoring under high-frequency read/write traffic; apply the same API design discipline to FastAPI GenAI services.
- Established **CI/CD workflows** (GitHub Actions / Jenkins) that reduced deployment time across staging and production; packaged and operated services with **Docker** on AWS (EC2, VPC, IAM, ECS, Lambda, S3, CloudWatch).
- Optimized **MySQL** queries and schema design for high-frequency operations; built React.js dashboards for real-time cloud monitoring.
- Collaborated with DevOps and infrastructure teams in Agile sprints; participated in code reviews and reliable release delivery.

GENAI PROJECTS (PRODUCTION-STYLE)

Agentic RAG Advisor — Lead Specialization Capstone (Regulated domain) Shipped · Oct 2026

LangGraph · LangChain · ChromaDB / FAISS · RAGAS · LangSmith · FastAPI · Docker · Kubernetes

Add public GitHub / demo URL before applying.

- Built an end-to-end **agentic RAG** advisor: document ingestion, chunking, embeddings, vector retrieval, prompt assembly, guardrails, and grounded generation.
- Orchestrated multi-step reasoning with **LangGraph** (state graphs, memory, tool use, persistence, human-in-the-loop, subgraphs) over a retrieval pipeline.
- Exposed capabilities via **FastAPI** microservices with REST endpoints, validation, structured errors, and basic observability.
- Instrumented workflows with **LangSmith** tracing; maintained a **RAGAS** evaluation harness for retrieval and answer quality.
- Packaged with **Docker**, deployed to **Kubernetes**, and automated build/test/image publish with GitHub Actions.

Additional GenAI Capstones Oct 2026

LangChain · CrewAI · Vector DBs · Open-source + API LLMs · FastAPI · Tool use

- Domain-aware LLM chatbot** — model switching between open-source (Mistral) and API LLMs with prompt control and quality tuning.
- RAG customer support assistant** — retrieval over internal knowledge bases (ingestion, embeddings, vector search, grounded answers).
- Agentic market-insight workflow** — multi-step agents with web search tools, routing, memory, and structured outputs.

GENERATIVE & AGENTIC AI SPECIALIZATION

Advanced Generative and Agentic AI — Naresh Technologies

Completed Oct 2026

- **Foundations:** Python for AI, neural networks, PyTorch, Transformer architecture (self-attention, multi-head & cross-attention, encoder–decoder), Hugging Face, multimodal LLMs, Ollama, prompt engineering & guardrails.
- **RAG systems:** chunking strategies, embeddings, ChromaDB, Pinecone, FAISS, Milvus, conversational RAG, RAGAS evaluation, Corrective RAG, Self-RAG, hybrid search, GraphRAG, Neo4j & Cypher.
- **Agentic AI:** LangChain, CrewAI, LangGraph (state graphs, workflow patterns, persistence, time travel, human-in-the-loop, subgraphs), MCP (FastMCP, langchain-mcp-adapter), A2A, n8n, LangSmith observability.
- **Optimization & serving:** PEFT fine-tuning (LoRA/QLoRA), quantization, vLLM, TensorRT-LLM.

ADDITIONAL DELIVERY

PULSE — Cloud Operations Platform

Arche Global

React.js · Node.js · Express.js · MySQL · AWS · Docker · RESTful APIs

- Architected a comprehensive cloud operations platform for hybrid environment management — the central tool for infrastructure visibility and incident response.
- Designed 15+ RESTful API endpoints and a MySQL-backed data layer with optimized queries across asset and ticketing modules; built real-time monitoring dashboards in React.js.

BIAL Plus — Internal Employee App (Android)

Production · Google Play

Kotlin · Jetpack Compose · Azure AD · Intune SDK · CI/CD

- Developed and shipped the official internal mobility platform for Bangalore International Airport employees, centralizing HCM, shift management, and HR services.
- Built ground-up in Kotlin/Jetpack Compose; integrated Azure AD SSO and Microsoft Intune (MDM/MAM); resolved critical auth blockers to support 24/7 uptime for mission-critical airport operations.

ArcheOne — AI-Powered Productivity Platform (Android)

Arche Global

Kotlin · Android SDK · RESTful APIs · Real-time Notifications

- Delivered end-to-end features (Business Card, SOS, Deskart, TravelDesk, Helpdesk) with real-time push notifications and optimized REST integrations across a wide device range.

EDUCATION & CERTIFICATIONS

Bachelor of Technology (B.Tech) in Engineering — Sri Venkateswara College of Engineering, Chennai

2021

- Advanced Generative and Agentic AI — Naresh Technologies · Completed October 2026
- AWS Certified Solutions Architect – Associate · Valid until March 2027
- APTECH Certified Python Developer

Languages: English (Fluent) · Hindi (Fluent) · Tamil (Native)

RELEVANT STRENGTHS FOR FULL-STACK GENAI ROLES

- **Agentic AI & LLM orchestration:** LangGraph, LangChain, CrewAI — reasoning, tool use, memory, and multi-step workflows.
- **RAG implementation:** ingestion, chunking, embeddings, vector search, grounded generation, and RAGAS evaluation.
- **Backend & API engineering:** FastAPI microservices for AI capabilities; prior production REST APIs at scale.
- **Containers & CI/CD:** Docker packaging, Kubernetes deployment, GitHub Actions pipelines.
- **Observability:** LangSmith tracing and evaluation harnesses for retrieval/answer quality.
- **Cloud & engineering practice:** AWS production delivery, Git workflows, code reviews, Agile collaboration; regulated-domain GenAI capstone.